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Baskaran Srirangam

SENIOR ENGINEERING LEADER | MECHANICAL DESIGN & STRUCTURAL ANALYSIS EXPERT | PROGRAM & PROJECT MANAGEMENT PROFESSIONAL| SUBJECT MATTER EXPERT-COMPUTER AIDED ENGINEERING-CAE-FEA/CFD/MBD/DEM

*Seasoned Mechanical Engineer with **20+ years of expertise** in **CAE-Structural Analysis, Finite Element Simulations, coupled structural-thermal-analysis**, CFD & product Engineering. Seeking a senior management or project management or Subject Matter Expert-CAE or consultant role. Aiming to leverage deep domain expertise in CAE, FEA, CFD, product Engineering, system Engineering, Customer Interfacing and Cross-functional project execution to drive innovation, improve product performance and lead high-impact engineering programs within a technology-driven organization.*

PROFILE SUMMARY

- Seasoned Mechanical Engineering Professional with **rich experience** in **Product Engineering**, **Electro-Mechanical Design, Finite Element Analysis (FEA)**, **Structural & Computational Fluid dynamics (CFD) analysis** with proven expertise in leading end-to-end design and CAE simulation projects
- **Several years of experience in Leading/ Managing aerospace, Automotive, Heavy Engineering & product Engineering,** development and CAE simulations program for global customer
- Adept at **technical program management**, client engagement, and cross-functional leadership, currently operating in a **Senior Manager**, handling **multidisciplinary teams and design-analysis integration**.
- **FEA Tools**: Hands on working experience of **ANSYS**, **Hypermesh**, **Abaqus**, **MSC** Nastran, **RADIOSS**, **LS-DYNA, ANSYS FLUENT (CFD) & SolidWorks flow simulations** for complex simulation tasks including stress, thermal & Flow, vibration, fatigue, and crashworthiness analysis. **CAD & Design Tools**: Hands on experience in SolidWorks and **NX/CATIA** for mechanical and structural design, DFMEA, and GD&T implementation.
- **Program/Project Leadership**: Proven track record of managing multiple R&D projects across global clients, overseeing delivery, schedules, technical risk management, resource planning, delivery governance, KPIs, SLAs
- **Product Lifecycle Ownership**: Extensive experience in supporting the full product development cycle — from requirement gathering, feasibility studies, detailed engineering, virtual testing, and prototype validation to manufacturing handover.
- **Technical Team Leadership**: Mentored and led high-performance teams of design & analysis engineers across mechanical domains including aerospace, automotive, and heavy engineering sectors, manage onshore/offshore resources
- **Customer-Facing Engineering Leadership**: Handled global clients in a **techno-commercial capacity**, supporting pre-sales, design feasibility, and solution architecture, drive account growth while aligning project delivery with client expectations.

CORE COMPETENCIES

- Finite Element Analysis (FEA)
- Structural & Thermal Analysis
- Mechanical Product Design
- ANSYS / HyperMesh / Abaqus / MSC Nastran/ LS DYNA/RADIOSS
- Computational Fluid Dynamics (CFD)
- CFD-ANSYS FLUENT/ SOLIDWORKS FLOW
- Design for Manufacturability (DFM)
- Computer-Aided Design (CAD)
- SolidWorks, NX, Creo & CATIA v5
- Simulation-Driven Design
- Project & Program Management (PMP)
- Engineering Change Management (ECM)
- Design Validation & Testing
- Root Cause Analysis (RCA)
- Product Lifecycle Management (PLM)
- Cross-Functional Collaboration
- Verification/ Testing

NOTABLE ACCOMPLISHMENTS

- Led **end-to-end CAE & product life cycle** projects including **new product development**, **re-engineering**, **sustenance**, and **compliance**, resulting in reduced **time-to-market** and enhanced **product reliability** by applying computer aided Engineering process.
- Good in conducting structural stress analysis using a hybrid approach of **FEM methods (Nastran/Patran/Hypermesh/Hyperworks)** and **classical hand calculations** to ensure structural integrity and regulatory compliance.
- Good in evaluating production deviations (concessions) and performing **Structural Repair Manual (SRM)** assessments for aerospace programs.
- Skilled in the technical review of **FEM models, validation reports, and the consolidation** of **Structural Analysis Reports** for aerospace projects
- **Spearhead the automotive CAE analysis of BIW, chassis, cab-structure, seating system, braking system, EPS systems /steering system, airbags, and engine components for** ensuring static stress, dynamics analysis and fatigue analyses (LCF/HCF), significantly improving structural durability
- Delivered **regulatory-compliant crash and safety simulations** (**FMVSS**, **ECE-R**, **AIS standards**) for **automotive seating systems**, ensuring compliance during **validation audits**. Good understanding of structural mechanics, transient dynamic calculations & various material models
- Executed complex **NVH** and **vibration analysis** (**modal**, **random**, **transient dynamic**, **complex eigenvalue**) using tools like **Abaqus**, **MSC Nastran**, **Ansys**, and **Radioss**, optimizing **structural integrity** and **passenger comfort**. Good in Vibration, acoustic and vibro-acoustic modeling & analysis
- Optimized **thermal** and **fluid flow designs** for **ECUs** and **electro-mechanical systems** using **CFD tools** (**SolidWorks Flow**), improving **system performance** and **thermal management** by over 25%. Good in multi-physics simulations, thermal and flow dynamics analysis

❖ CAE enabled Designed cost-effective components using **sheet metal**, **plastics**, and **castings** with strong knowledge of **DFM**, **DFA**, and **rapid prototyping**, achieving up to 20% **manufacturing cost savings**. Good in ASME, ASTM, and ISTA standards

❖ Good in **2D/3D CAD models** with **GD&T**; **DFMEA**, **VA/VE**, **APQP**, and **V&V processes**, robust and manufacturable designs.

Automotive Product Engineering & CAE Analysis | Domain: Passenger & Commercial Vehicles | Clients – ARAI, TAFE, TRW, GM, Mazda, ICML, Eicher

Key Contributions: CAE Structural analysis of chassis & braking system, steering system, seating system, airbags for Tier 1 supplier (USA & Germany)

Vehicle body CAE structural analysis: FEA of cabs and BIW structures for commercial buses, trucks, SUVs and automotive vehicle

Off-highway vehicle: NPD/NPI of 40 HP Farm tractor design and structural analysis. Re-Engineering of material handling crane & port cranes (RTG)

Aero-Structures| Domain: Aerospace | Metallic and composite structural analysis & Crashworthiness

Key Contributions: Structural analysis of wing composite structure, Bird impact analysis of leading edge and crashworthiness of seating system, CAE enabled design and analysis of composite structure, fatigue and fracture analysis of composite and metallic structure & knowledge of FAA & EASA regulations

Industrial system & Mining Industry: Structural analysis of elevators sub-system, MCC structure and Electro-Mechanical power system CFD analysis

Worked on Mining innovation projects for critical structural & thermal simulation for ore handling systems (DEM), ship loader decks- imaging systems thermal simulation and vehicle laser enclosures design and development for ore explorations **Plant Engineering:** NPD/NPI solar dish CAE enable design and CAE

Medical Device Design, Development & CAE Validation | Domain: Electro-mechanical Medical Products | Global Clients – USA, Europe, Japan

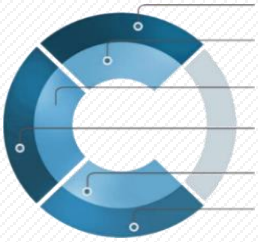
Key Contributions: Led CAE-driven design and structural analysis for a range of electro-mechanical medical devices including **in-vitro diagnostics SPO2 analyzers, CGMS systems, blood-urine analyzer, blood apheresis device, bone plate, medical scale, ventilator sub-system and endoscopy flushing devices.**

Drug delivery device FE analysis and CAE enable design & development, verification test case and failure analysis

Consumer Product Design & CAE Simulation | Domain: Consumer Electronics & Appliances | Clients – Philips, BSH, Arçelik

Key Contributions: Spearheaded NPD & NPI, CAE-SPOC for consumer lifestyle products across Philip's global technical centers (Netherland, USA, APAC)

SKILL MATRIX



Progressive Leadership
Technology & Innovation
Data Driven Decision Making
Problem-solving and Result-oriented
Analytical Thinking and Critical Analysis
Global Mindset

EDUCATION

- **1998:** Masters in Engineering (M.E.) in Mechanical Engineering (Design Eng) Gov't College of Engineering Pune, India
- **1995:** Bachelor of Technology (B.Tech) in Mechanical Engineering from Pondicherry University, Pondicherry, India

ON-SITE EXPERIENCE

- **June-2026- German Chancenkarte Visa (Tentaively Visa expected)**
- **Mar;23:** Onsite to **Germany and Czechia medical device plant** for KT and product training -2 weeks
- **Nov'09 – Jan'10: Philips Oral Health Care** in Seattle WA, USA for Technology and function creation
- **Nov'03: Toro** Company for 3 weeks in Minnesota, USA
- **Dec'01 – Feb'02:** Participated in **TRW Brakes at Koblenz, Germany** for analysing brakes systems in NVH domain

TECHNICAL SKILLS

- **FEA Software:** Expert in Hypermesh, Opistruct, LS Dyna, RADIOSS, MSC Nastran, Abaqus CAE/ Simulia, ANSYS Mechanical, ANSA, JMP-DOE
- **CFD Software:** Solid Works Flow Simulation,ANSYS Fluent, Star-CCM
- **MBD:** Altair Motion solve, SolidWorks motion analysis, MSC Adams
- **CAD Software:** SolidWorks, NX CAD, Creo (Pro-E) & CATIA v5
- **Programing software :** Python, C,

LANGAUAGE

English -Fluent - Read, Write & Speak
German- (~A1/A2Pursuing)- Basic/Intermediate- Read, Write & Speak

WORK EXPERIENCE

Aug 2022 – Sep 2025| Medmix (Haselmeier India Pvt. Ltd) Bengaluru, India | Senior Manager-CAE

Sep 2010 – Sep'22 Capgemini Technology Services India Ltd., Pune, India | Manager

Jul 2006 – May'10 | Cyient (Formerly Infotech Enterprises Limited), Hyderabad | India, Assistant Project Manager

Nov 2004 – Jun'06 | Eicher Engineering Solutions Ltd (A Unit of Eicher Motors Ltd.), Gurgaon, India | Project Leader (Manager)

Feb 2004 – Aug'04 | Quantech Global Engineering Services Ltd, Hyderabad, India | Senior Project Engineer

Jul 2000 – Jan'04 | Satyam – Venture Engineering Services Pvt. Ltd., Hyderabad, India | Senior Analyst

Jun 1999 – Jul'00 | Tractors and Farm Equipment Ltd. (TAFE), Chennai, India | R&D Member – New Product Design

Dec 1998 – March'98 | JJ college of Engineering and Technology, Trichy, India | Lecturer in Mechanical Engineering

ANNEXURE -PROJECTS UNDERTAKEN

Automotive Product Engineering & CAE Analysis | Domain: Passenger & Commercial Vehicles |

Key Contributions:

- Executed end to end structural simulations for Automotive subsystems: full vehicle BIW structure, chassis, commercial vehicle cab, seating system, braking system, EPS/ EPAS steering system, airbags, and engine components.
- Delivered CAE validation and engineering calculations for NVH, crash safety, and occupant protection systems.
- Offshore delivery with TRW Technical Centers (Germany & USA) for steering and braking system analysis.
- Conducted simulations for vehicle dynamics, subsystem testing, and validation using industry-standard tools.
- Drove design verification and virtual validation (V&V) for off-road and heavy-duty vehicles including tractors and cranes.

Automotive Design, FEA & Crashworthiness Programs

Stress and Durability analysis: Off-Highway, HCV, LCV, BIW & Subsystems

- Led design and structural FEA of cabs, chassis systems, and BIW structures for commercial buses, trucks, and SUVs.
- Delivered stress, stiffness, and fatigue analysis to meet global safety standards and homologation targets.
- Designed and validated 40 HP tractor platform for TAFE Ltd targeting the Indian market.
- Conducted fatigue and stress analysis of braking components (caliper, carrier, knuckle, disc & drum brakes) for TRW, Germany & USA.
- Executed FEA of steering and suspension systems, including control arms and multi-spoke steering wheels.
- Supported IC engine design with structural and cooling cavity simulations.

Automotive Crashworthiness & Safety Analysis

- Performed full vehicle crash analysis: frontal, side, rear, and rollover in accordance with FMVSS and ECE-R norms.
- Delivered crash safety compliance for LCV and HCV cabins under ECE R29/Indian standards for Eicher Motors.
- Led aircraft and automotive seating system crash validation using LS-DYNA.
- Conducted simulations for head impact, side door intrusion, bumper crash, and airbag deployments.

Automotive NVH & Multi-body Dynamics

- Delivered NVH analysis for vehicle subsystems and full platforms; optimized brake squeal and frequency response (FRF).
- Performed modal, eigenvalue & vibrational analysis for brake systems and seating structures using Nastran, Hypermesh.
- Designed and analyzed muffler systems and predicted shell noise using FEM/BEM techniques.
- Executed multi-body dynamics simulations with Altair Motion, SolidWorks Motion, and MSC Adams.

Complex computation fluid Dynamics (CFD) flow and thermal simulation

- Delivered CFD Thermal management and thermal optimization for Electro-Mechanical products
- External and internal flow simulation, 1D/3D flow & thermal simulation for medical and industrial/ Mining applications
- Steady, transient, turbulent modeling, viscous modeling, multi-physics simulation for range simple to complex simulations
- CFD meshing, simulation, interpretation of fluid dynamics results and optimal solutions
- Performed range of electronic cooling simulation for medical and industrial products to improve thermal performance

Aerospace Structures & Crashworthiness

Aircraft Metallic & Composite Structures

- Performed comprehensive structural stress analysis for aerospace components, utilizing Nastran, Hypermesh and Patran for global FE modeling while validating results through classical hand calculations
- Composite material FE structural analysis and simulations of aero-structure of aircraft. Implemented standardized procedures for FEM model descriptions and validation, improving the consistency and accuracy of structural reports across the engineering team.
- Leveraged the Structural Repair Manual (SRM) to perform detailed repair assessments, incorporating relevant references and technical data into formal analysis documentation.
- Evaluated production deviations and non-conformities, providing response stress-related justifications to maintain production timelines.
- Executed bird impact simulations on leading edge of aero-wings and failure evaluations
- Led nonlinear analysis of crash events for aircraft seat structures and safety systems using explicit analysis
- Applied first-principles design/ hand calculations and composite material failure analysis
- Knowledge in aerospace standards including AS9100/AS9145 (APQP/PPAP), MIL-STD, EASA and FAA standards
- Knowledge in Aero-structure design, aero-dynamics CFD, composite material design, manufacturing process, Fatigue & fracture mechanism and system Engineering

<u>Heavy Engineering, Mining & Plant Engineering</u>

Engineering Design & Structural Validation

- Managed design and structural analysis for ore handling systems, marine decks, port cranes, elevators, and solar dishes.
- Delivered thermal-structural analysis for ship loader decks- imaging systems, and vehicle laser enclosures
- Re-engineered Rubber Tired Gantry (RTG) cranes for improved container handling efficiency for Europe customer
- CAE enabled Design and analyzed IMT cranes for US market, addressing stability and operational performance.
- NPD and structural analysis of solar dish for power generation projects

Medical Device Design, Development & CAE Validation | Domain: Electro-mechanical Medical Products | Global Clients – USA, Europe, Japan

Key Contributions:

- Program and project management of design and structural analysis for a range of electro-mechanical medical devices including **in-vitro diagnostics, SPO2 analyzers, CGMS systems, blood-urine analyzer, blood apheresis device, bone plate, medical scale, ventilator sub-system and endoscopy flushing devices**.
- Delivered CAE analysis for new product development (NPD) and sustenance engineering across global compliance standards.
- Executed value engineering and re-design programs to optimize cost and performance.
- Designed drug delivery pens and glucose monitoring systems for diverse regulatory markets (USA, EU, APAC).
- Carried out CAE analysis of drop, shock, and vibration test simulations per IEC and ISTA standards for packaging validation.
- Leveraged design optimization and weight reduction, ensuring structural integrity and regulatory compliance.
- Product documentation including **DHF, DMR, risk management, system Engineering and qualification IQ, OQ, PQ**

Consumer Product Design & CAE Simulation | Domain: Consumer Electronics & Appliances |

Key Contributions: **Global leader in consumer life style product -Netherland, Austria, USA, Singapore, ATC & global Tech center for 3 yrs.**

- Spearheaded NPD & NPI for consumer lifestyle products across global technical centers.
- Delivered structural and thermal simulations for products like shavers, epilators, irons, and garment care appliances.
- Performed product drop, impact, vibration test analysis for handheld devices to ensure reliability and durability.
- Acted as CAE SME/ SPOC for global COE for all simulation activities, coordinating deliverables between India and international teams.
- Applied FEA and CFD tools to enhance design feasibility and system-level performance.
- Supervised multidisciplinary teams in executing structured problem-solving, mold flow analysis, and material behavior studies

<u>Program & People Management</u>

Strategic & Technical Leadership

- Strategic leadership, technology development, research and development and innovation management
- Several years of experience in project management, managing cross-functional team & engineering programs across domains.
- Managing various project, budgeting, project planning, costing, scheduling, resource allocation, and risk mitigation for global clients.
- Delivered PMO dashboards, utilization reports, and stakeholder communications for top management.
- Managed RFI/RFQ preparation, customer specs, and proposal development.
- Facilitated client engagement, resolving cross-disciplinary issues and enhancing satisfaction.

Team Management & Organizational Development

- Led hiring, competency mapping, training, appraisals, and performance evaluations.
- Competency development for mechanical design, new engineering services capability building and contribute digital transformation
- Methodology development, Proof of concept and feasibility evaluation
- Adhering QMS and quality assurance processes across engineering teams.